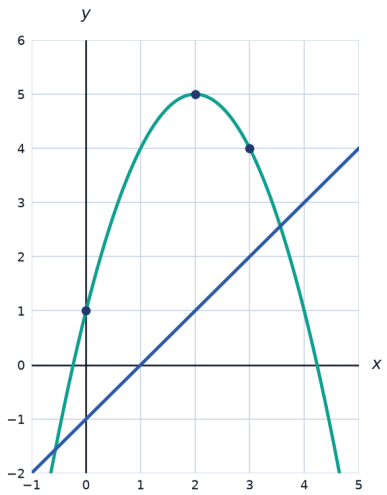
Functions & Transformations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Transformations of y=x

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(5)$.

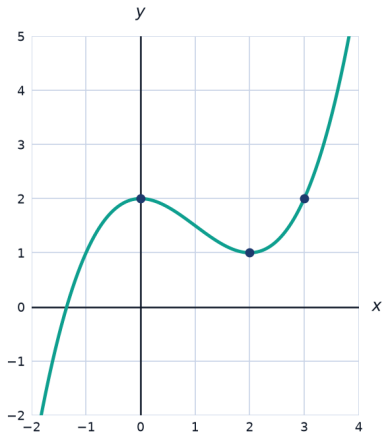


2 [Foundational] • Target: Transformations of y=x

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: Slope-intercept form

For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(2)$.



4 [Foundational] • Target: Slope-intercept form

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

5 [Foundational] • Target: Slope-intercept form

A data set recorded for slope-intercept form consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

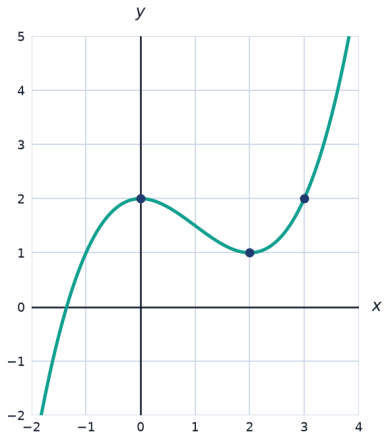
© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Slope-intercept form

For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(3)$.

7 [Foundational] • Target: Slope-intercept form

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Slope-intercept form

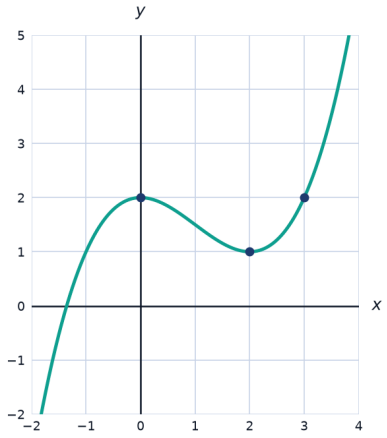
For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(4)$.

9 [Intermediate] • Target: Slope-intercept form

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

10 [Intermediate] • Target: Slope-intercept form

For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(5)$.



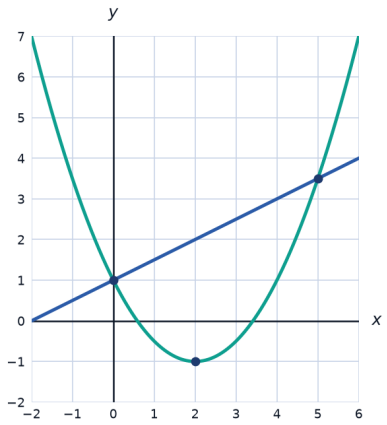
© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Slope-intercept form

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

12 [Intermediate] • Target: Standard form

For the model $f(x) = 4x + 7$ in standard form, evaluate $f(2)$.



13 [Intermediate] • Target: Standard form

A data set recorded for standard form consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Standard form

For the model $f(x) = 4x + 7$ in standard form, evaluate $f(3)$.

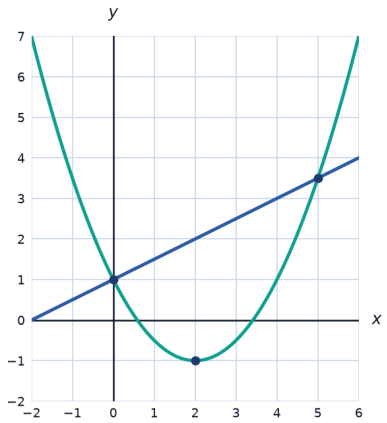
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Standard form • Cross-Lesson: 9.01 Radicals & Exponents

For the model $f(x) = 4x + 7$ in standard form, evaluate $f(4)$.

16 [Challenging] • Target: Standard form • Cross-Lesson: 1.01 Linear Equations & Lines

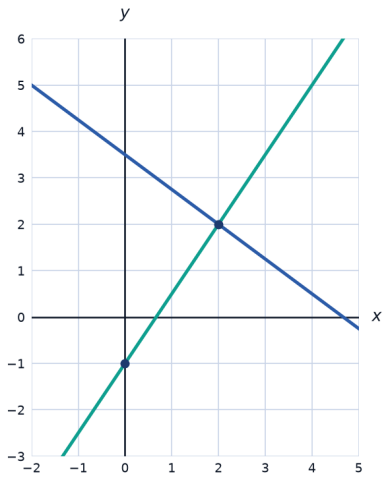
For the model $f(x) = 4x + 7$ in standard form, evaluate $f(5)$.



© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Point-slope form • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 8$ in point-slope form, evaluate $f(2)$.



18 [Challenging] • Target: Point-slope form • Cross-Lesson: 11.01 Statistics & Data Analysis

A data set recorded for point-slope form consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Point-slope form • Cross-Lesson: 2.01 Quadratic Functions

For the model $f(x) = 5x + 8$ in point-slope form, evaluate $f(3)$.

20 [Challenging] • Target: Point-slope form • Cross-Lesson: 3.01 Exponents & Polynomials

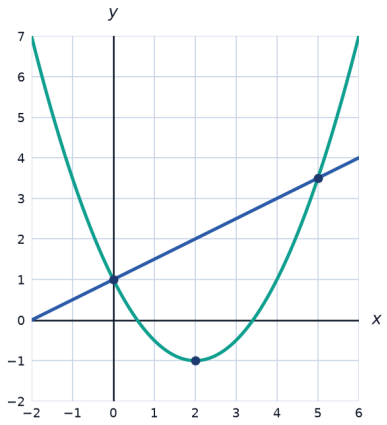
For the model $f(x) = 5x + 8$ in point-slope form, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Review: Slope as rate of change

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(2)$.



a. 13

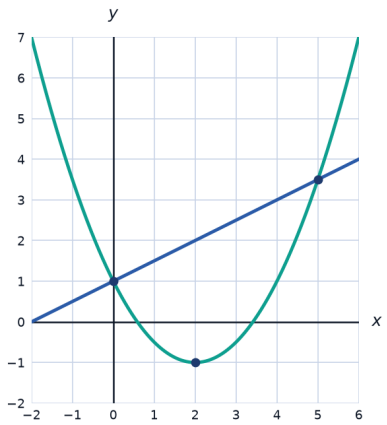
b. 18

c. 10

d. 17

22 [Foundational] • Target: Review: Slope as rate of change

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

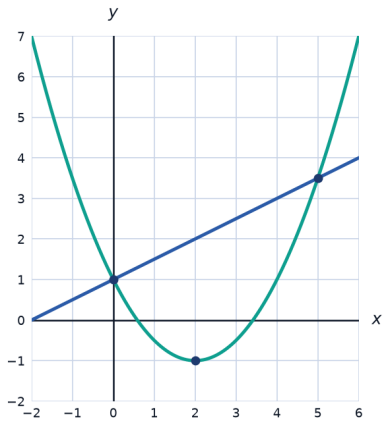
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Review: Slope as rate of change

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(5)$.



a. 28

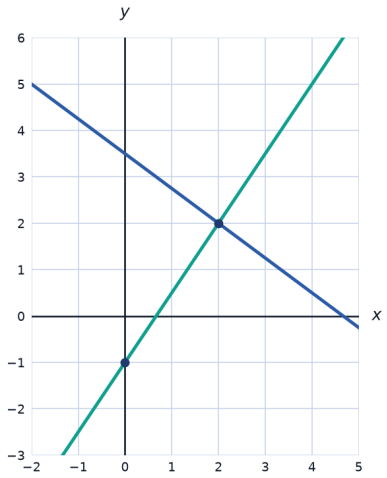
b. 33

c. 25

d. 32

24 [Foundational] • Target: The slope formula

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(2)$.



a. 16

b. 22

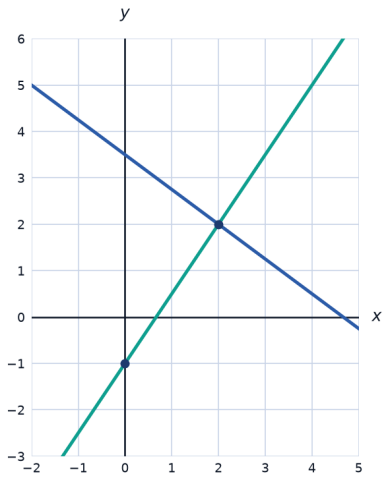
c. 12

d. 20

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: The slope formula

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

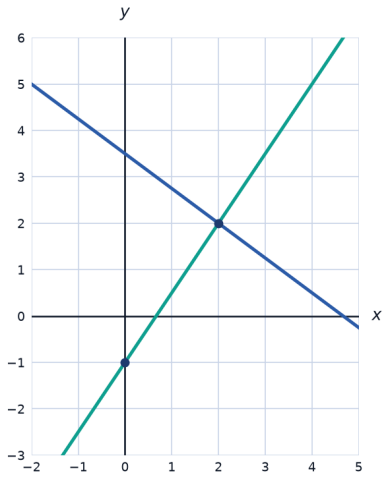
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: The slope formula

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(5)$.



a. 34

b. 40

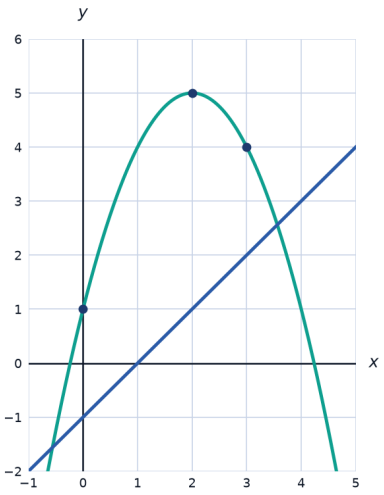
c. 30

d. 38

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Transformations of y=x

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(2)$.



a. 9

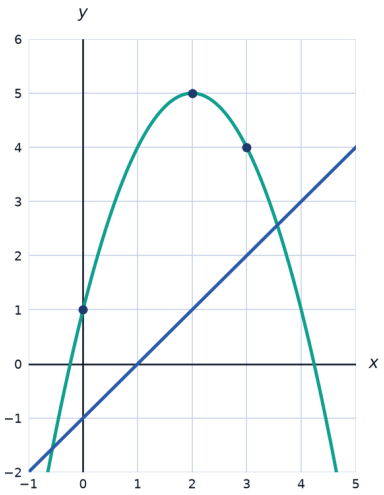
b. 11

c. 4

d. 13

28 [Foundational] • Target: Transformations of y=x

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Review: Slope as rate of change

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Review: Slope as rate of change

A data set recorded for review: slope as rate of change consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

31 [Intermediate] • Target: Review: Slope as rate of change

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(3)$.

a. 18

b. 23

c. 15

d. 22

32 [Intermediate] • Target: Review: Slope as rate of change

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(4)$.

a. 23

b. 28

c. 20

d. 27

33 [Intermediate] • Target: Review: Slope as rate of change

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Review: Slope as rate of change

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

35 [Intermediate] • Target: The slope formula

A data set recorded for the slope formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

36 [Intermediate] • Target: The slope formula

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(3)$.

a. 22

b. 28

c. 18

d. 26

37 [Intermediate] • Target: The slope formula

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(4)$.

a. 28

b. 34

c. 24

d. 32

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The slope formula • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: The slope formula • Cross-Lesson: 2.01 Quadratic Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

40 [Challenging] • Target: Transformations of y=x • Cross-Lesson: 3.01 Exponents & Polynomials

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

41 [Challenging] • Target: Transformations of y=x • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(3)$.

a. 11

b. 13

c. 6

d. 15

42 [Challenging] • Target: Transformations of y=x • Cross-Lesson: 6.01 Systems & Inequalities

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(4)$.

a. 13

b. 15

c. 8

d. 17

43 [Challenging] • Target: Transformations of y=x • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Slope-intercept form • Cross-Lesson: 8.01 Factoring Polynomials

A data set recorded for slope-intercept form consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

45 [Challenging] • Target: Standard form • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for standard form consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

© ClipSAT — clipsat.org | All Rights Reserved